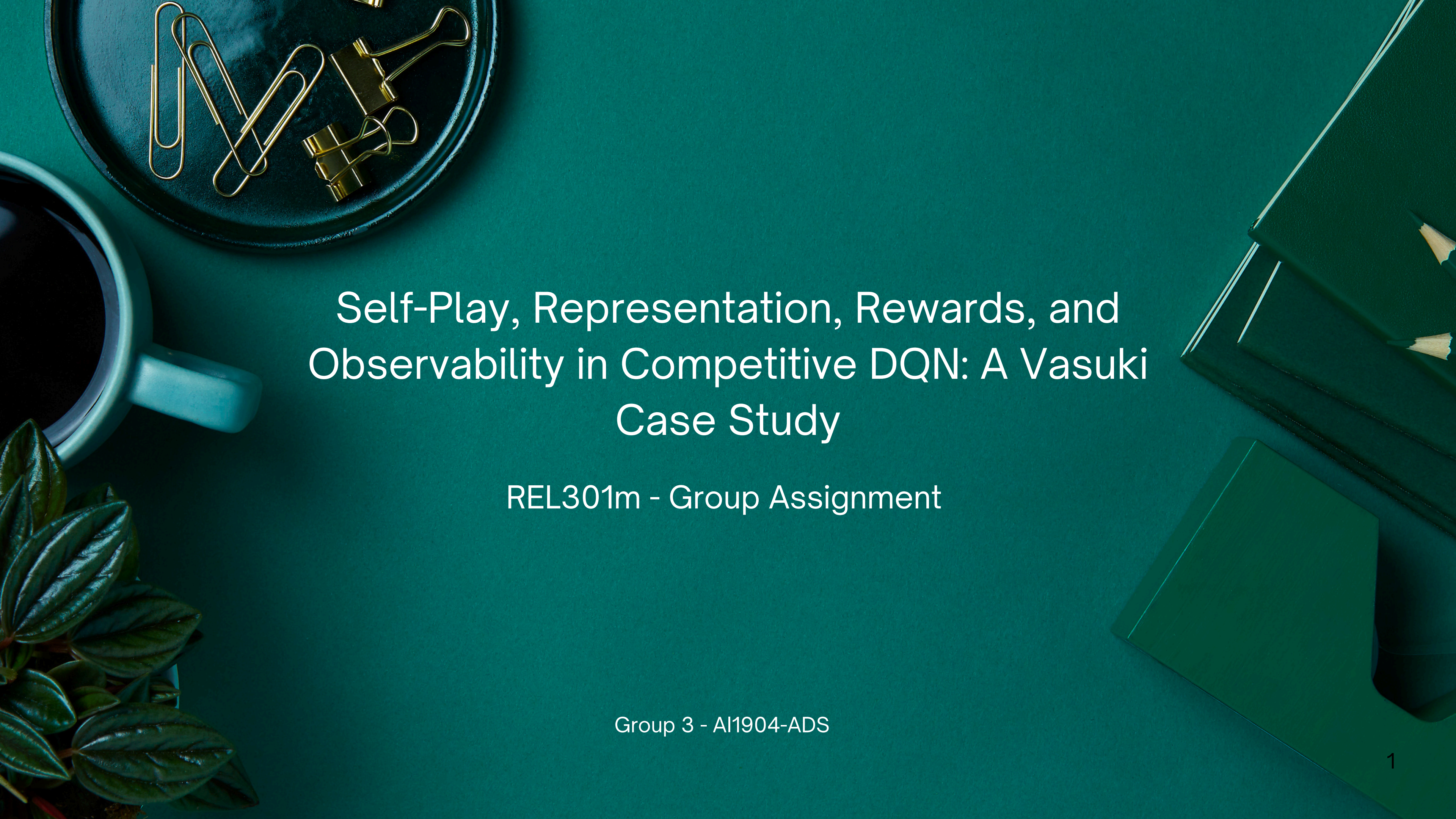




Self-Play, Representation, Rewards, and Observability in Competitive DQN: A Vasuki Case Study

REL301m - Group Assignment

Group 3 - AI1904-ADS

Table of Contents

Problem Framing

Direction 1 · Self-Play League

Direction 2 · Representation

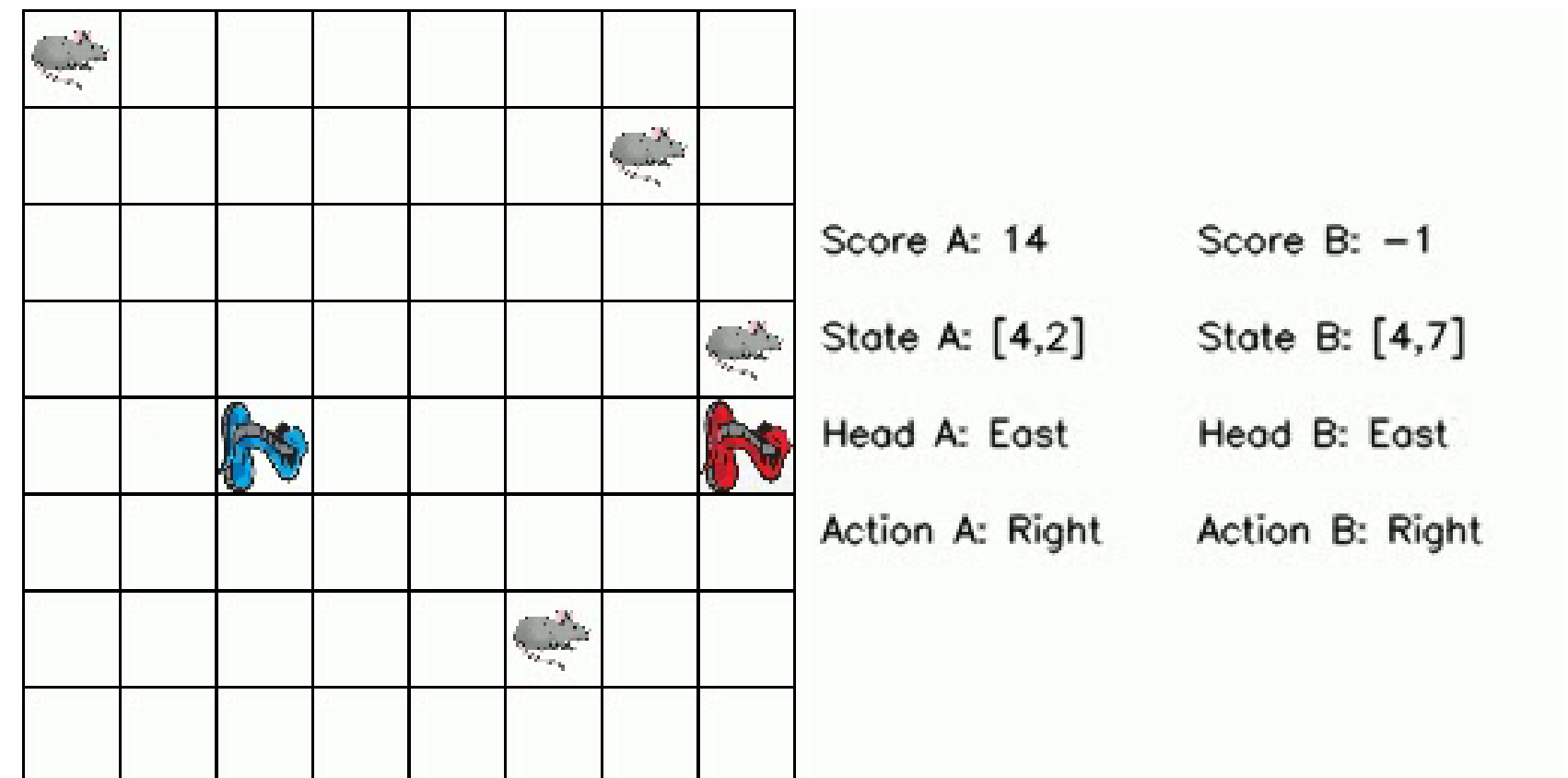
Direction 3 · Reward Shaping

Direction 4 · Observability

Conclusion

Vasuki: Competitive DQN in a Two-Snake Grid World

- Vasuki is a two-snake competitive grid environment
- Both snakes race for food while avoiding illegal moves and collisions
- Each agent's result depends on both its own policy and the opponent it faces
- This makes evaluation noisy: a good-looking policy may only be good against a weak opponent
- Results are directional at this scale, not final claims



One Environment, One Shared DQN Setup

Environment

8x8 board
Two snakes compete for food
Actions: left, right, forward
Episode length: 100 steps

DQN Training

Algorithm: DQN
Library: Stable-Baselines3
Discount: $\gamma = 0.95$
Training: 100,000 timesteps
Seeds: 2

Evaluation

100 evaluation games
Main baseline: RandomOpponent
Compare each method to baseline
Do not rank all directions directly

Baseline : Against RandomOpponents

- RandomOpponent has no policy learning, no memory, and no adaptation
- It samples actions uniformly, so it gives a simple lower-bound opponent
- We use it to check whether the trained DQN learned basic survival and food-seeking

Score A: 19	Score B: -72
State A: [4,1]	State B: [2,1]
Head A: North	Head B: North
Action A: Right	Action B: Left

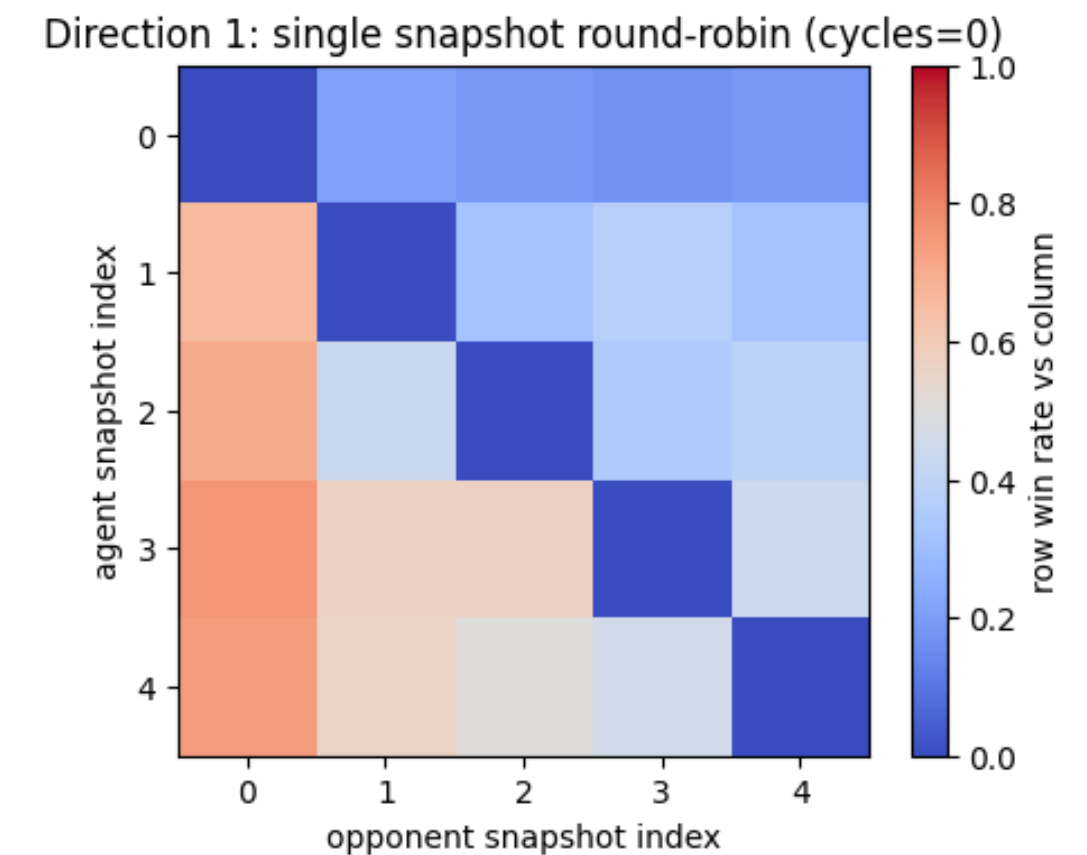
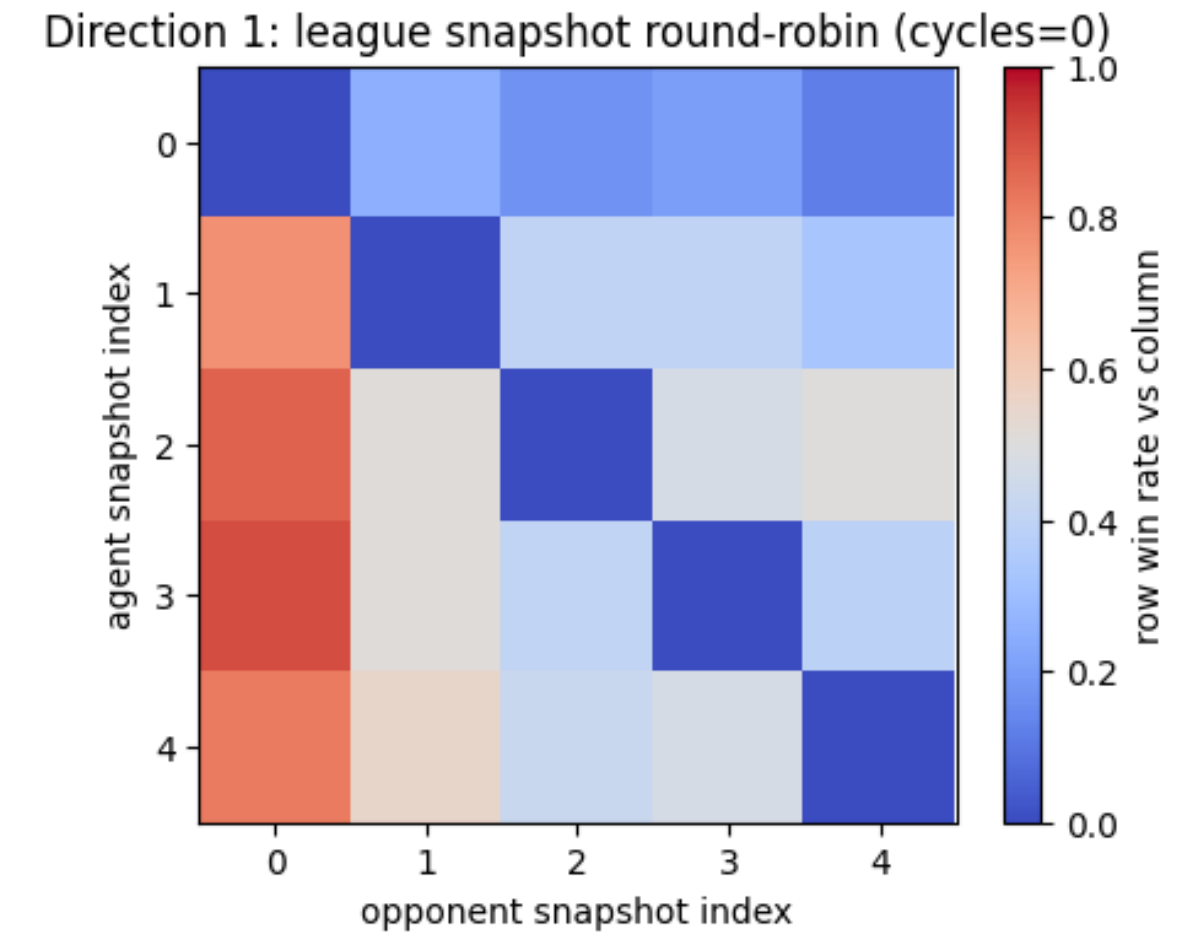
D1 Method: Single Snapshot vs League Self-Play

- Question: can opponent diversity make self-play more stable?
- Single snapshot: learns against one frozen past policy
- League: learns against a small pool of recent frozen policies
- Both use the same DQN budget and refresh schedule
- Evaluation checks baseline skill, head-to-head play, and snapshot consistency



D1 Result: Self-Play Pattern

- Single and league both improved from early penalties to stable positive reward
- Both passed the shared baseline evaluation
- Snapshot round-robin found cycles = 0
- Saved policies followed a consistent ordering instead of rock-paper-scissors behavior
- Takeaway: this run showed stable self-play dynamics; league remains a promising extension for larger-scale tests



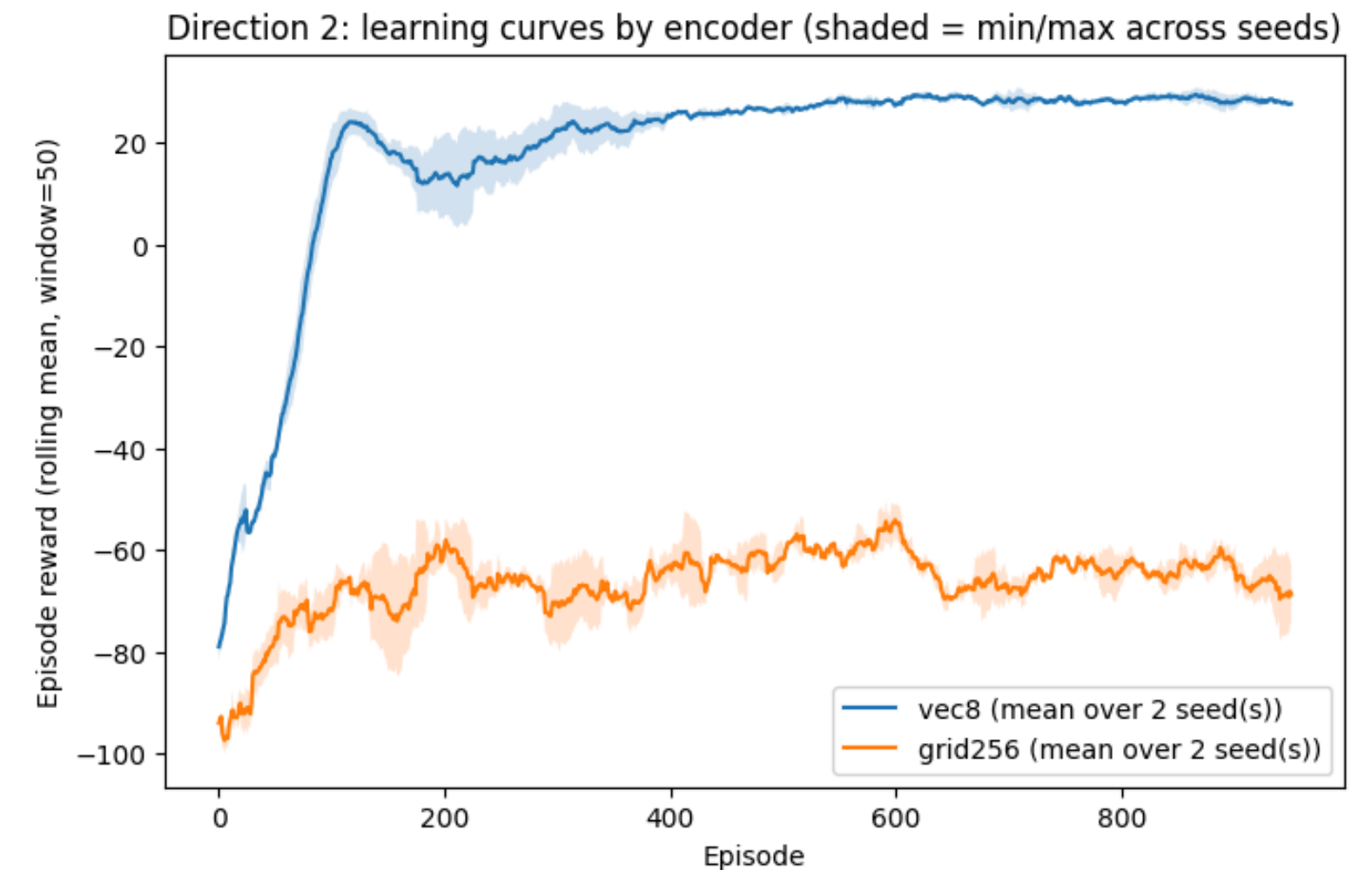
D2 Method: Hand-Crafted Vector vs Full Grid

Question: which observation representation learns faster and transfers better?

- vec8: compact 8-number hand-crafted state
- grid256: flattened 4-channel 8×8 board
- Both use the same MLP DQN
- Transfer test: vec8 trained at n=8, evaluated at n=12 without retraining

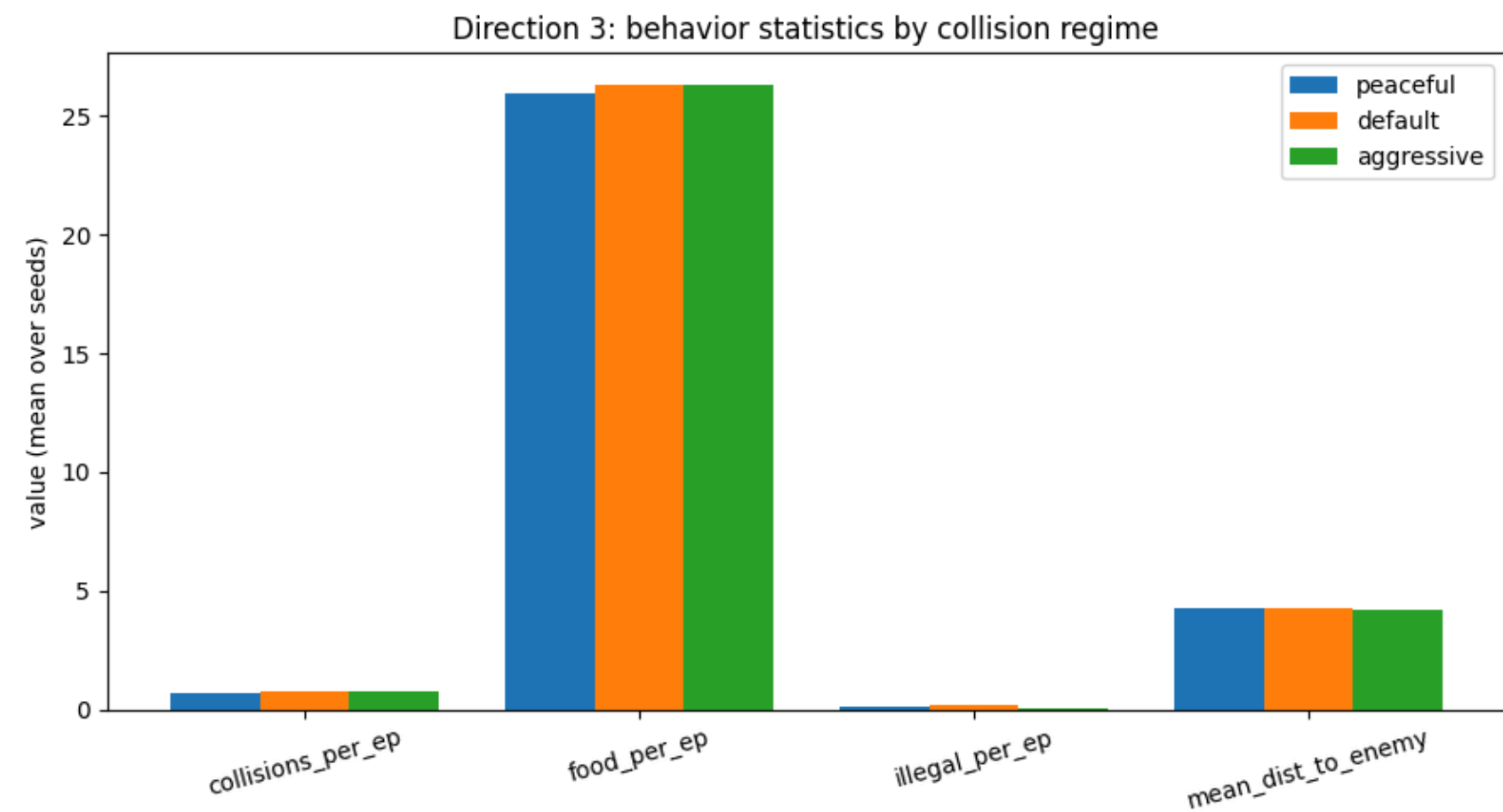
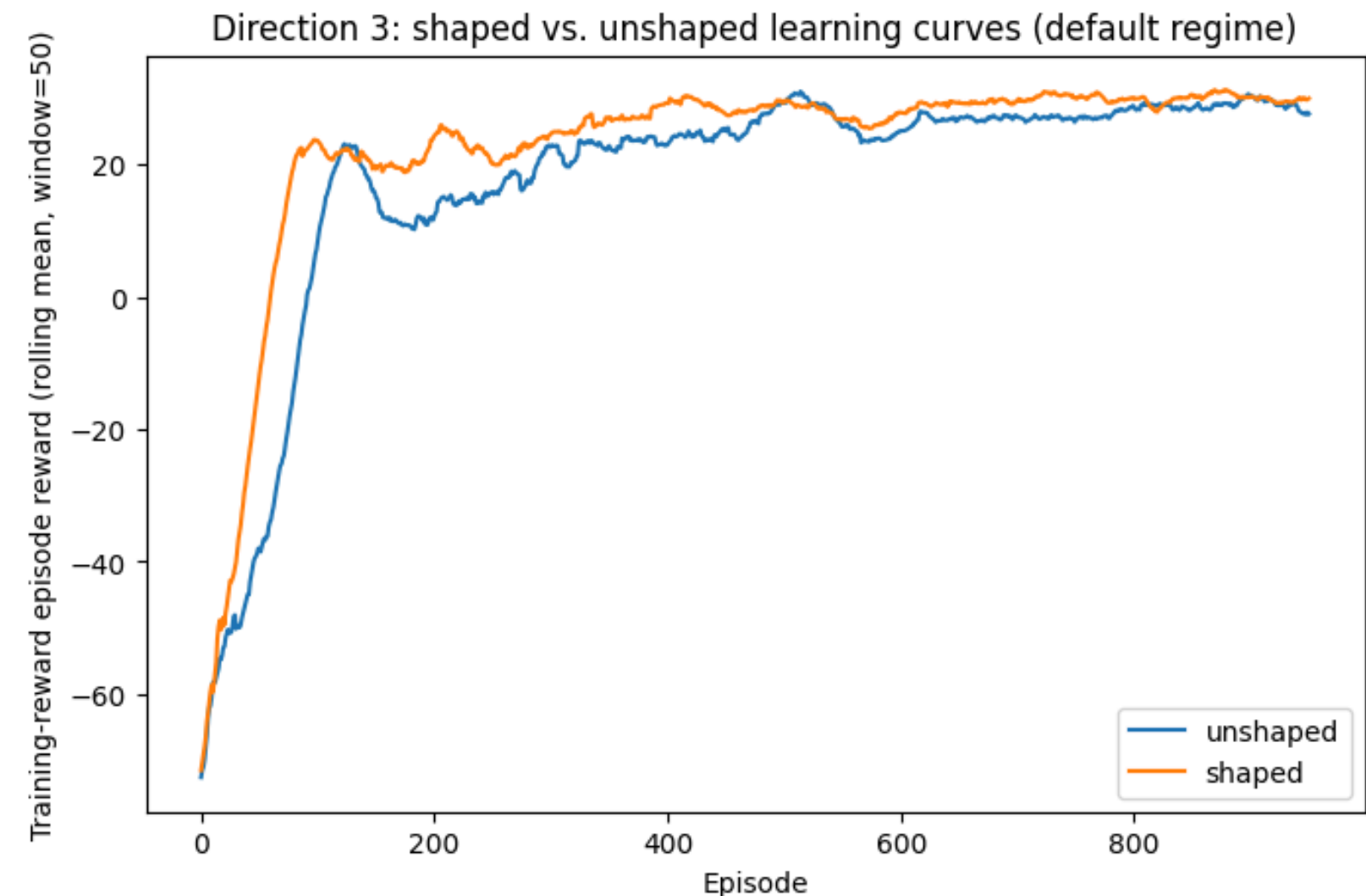
D2 Result: Compact Features Win Here

- vec8 reached 100% baseline win rate for both seeds
- grid256 reached 84% and 86% at n=8
- vec8 transferred zero-shot to n=12 with 100% win rate
- Fixed input size makes vec8 naturally portable across board sizes
- Takeaway: compact representation was the strongest result in this study



D3 Result: Reward Variants Reached Strong Baseline Performance

- All collision regimes achieved strong baseline performance
- Food collection stayed consistent across variants
- Illegal moves remained low
- Collision behavior stayed controlled across reward settings
- Shaped and unshaped agents both reached near-perfect final performance
- Takeaway: reward design is ready for deeper behavior testing against structured opponents



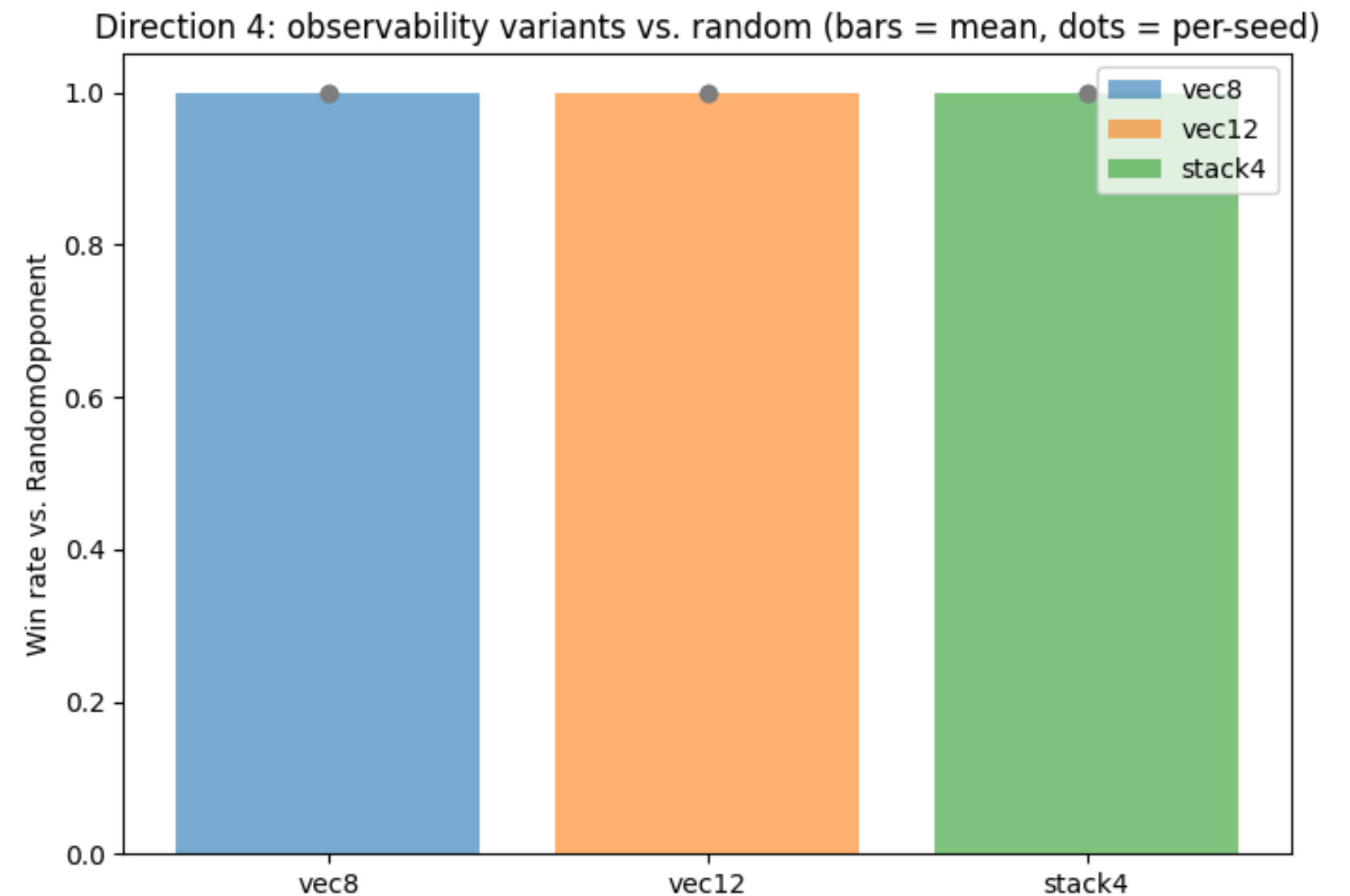
D4 Method: Observability and Elo

Question A: does missing enemy heading hurt performance?

- vec8: enemy position only, no enemy heading
- vec12: adds enemy heading as 4-value one-hot
- stack4: stacks four recent vec8 frames to infer motion
- Question B: use Elo round-robin when RandomOpponent win rate saturates

D4 Result: All Variants Beat RandomOpponent

- vec8, vec12, and stack4 all reached 100% baseline win rate
- Extra heading and short history both produced viable agents
- Elo places trained agents above the baseline opponent
- Agent-vs-agent ranking gives a stronger comparison path than baseline win rate alone
- Takeaway: observability testing should continue with structured opponent behavior



Conclusion & Future Works

- D1: self-play snapshots formed a stable ordering; no cycling appeared
- D2: vec8 gave the clearest gain: strong learning and zero-shot transfer to n=12
- D3: reward variants all reached strong baseline performance, with behavior staying controlled
- D4: all observability variants reached strong baseline performance; Elo adds a richer comparison layer
- Common insight: baseline testing confirms competence; next-stage evaluation should use more structured opponents
- Next: larger round-robin, more seeds, more games, and fuller behavior analysis

